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VALUATION TECHNIQUES

One of the most important skills of an entrepreneur or an investor is the ability to value an opportunity. These skills are critical when purchasing, growing or selling an ongoing business or, alternatively, starting a new one. Valuations, however, are more of an art than a science. Aspiring practitioners should therefore develop this art by first familiarizing themselves with the different valuation techniques.

Valuation techniques focus on a specific part of a company's financial performance. Three categories of techniques exist:

- balance sheet analysis,
- income statement multiples, and
- discounted cash flows.

Rather than yielding a single estimate of a company's worth, these techniques will likely yield a range of values that are always a function of many assumptions. This range will provide a skillful negotiator with a reasonable negotiating "boundary;" a final price will emerge after considering many financial and non-financial factors, including the valuation results. It should be noted that due to disparities in the assumptions, reasonable differences in perceptions of worth could arise among several parties even though each has used the same valuation technique.

This teaching note uses *Paint-Pen*, a case study of a company that is being sold, to show a realistic application of each valuation technique. Although Paint-Pen represents an example of a small and profitable company, the valuation techniques that will be reviewed in this note extend to large companies as well as unprofitable companies that forecast profitability.

It is not necessary to read the *Paint-Pen* case to understand the following review. The reader should only understand that in *Paint-Pen*, Mr. Hammer, a buyer, has been given five days to submit a bid to purchase Paint-Pen, a paint distribution company. Since the seller has imposed several restrictions on whom Mr. Hammer can contact to learn more about the company, very little is known about the operations other than the financial statements. Like in any valuation exercise, several "reasonable" assumptions must be

This note was prepared by Nick Mansour and Alexander Tauber under the supervision of H. Irving Grousbeck, Consulting Professor of Management, and Charles A. Holloway, Kleiner, Perkins, Caufield and Byers Professor of Management, Stanford University Graduate School of Business.

This note makes many references to the Paint-Pen case, HBS 9-898-156.

made in order to apply each valuation technique. These assumptions do not reflect access to any supplemental information.

BALANCE SHEET VALUATIONS

Balance sheet valuations attempt to calculate the worth of the underlying assets of a business as recorded in the accounting statements. According to generally accepted accounting principles, this is the remainder of total assets less total liabilities. Three applications of this technique are commonly used:

- book value,
- adjusted book value, and
- liquidation value.

BOOK VALUE

Book value is the simplest valuation technique. It is the assets minus the liabilities on a company's balance sheet. In 1996, Paint-Pen's total assets equalled \$4,118,110 and its total liabilities totalled \$936,460, resulting in a book value of \$3,181,650. As expected, this was exactly the value of "total capital stock and surplus". For a variety of reasons, the book value of an organization's assets rarely equals the true market value of the corresponding assets. Book value is therefore of limited use and rarely used to determine the final price of a transaction.

ADJUSTED BOOK VALUE

Adjusted book value is an attempt to reconcile the accounting values with the market values by assuming that the company will continue operating as an ongoing concern and approximating the replacement value of each asset and liability. Appraisals provide accurate estimates of the market values. The increased accuracy of these appraisals, however, do not justify the time and money involved with these efforts. It is therefore common to make rough estimates of the worth of each asset and liability using the practitioner's best judgment.

Assets and liabilities can be separated into four groups.

- Cash and cash equivalents. These assets are normally carried on the balance sheets at their market values – they are most often left unadjusted.
- Non-cash tangible assets. These assets include inventories, prepaid expenses, equipment, land, buildings, notes receivables, and accounts receivables. Inventories are affected by the likelihood of product obsolescence and the historical trend in raw material prices. Generally, the book value of inventories is overstated. Prepaid expenses should be changed to show their current value and the likelihood that the company will use them. The book value of equipment, land, buildings, and other similar assets is the historical purchase price less the corresponding depreciation or amortization. Depreciation and amortization schedules required by generally accepted accounting principles often do not reflect the economic life of these assets

and the book value therefore does not reflect their true replacement costs. Typically, depreciation schedules for assets employed in productive capacities (i.e., manufacturing equipment) are over shorter periods of time than their productive lives. Thus, these assets are often worth more than their corresponding book values. In contrast, general office equipment (furniture, computers, etc.) is often carried at values above its replacement cost. Its replacement cost is therefore normally estimated by discounting the corresponding value. The practitioner must recognize that there are no hard rules and that there may be specific situations where the productive assets are worth less than their book values, and the general office assets are worth more. Careful use of good judgment in each situation is a must. Notes and accounts receivables need to be adjusted for the likelihood that the company will get paid for them. By examining the historical allowances for doubtful collections in relation to the actual write-offs, anyone can estimate whether accounts receivables are under- or over-valued. Notes receivables can be adjusted using information about the creditworthiness of each debtor.

- Intangible assets. These assets include patents, non-competition covenants, license agreements, and goodwill. Patents, non-competes, and licenses should be adjusted to reflect the value they provide during their remaining life. Goodwill is usually assumed to be of no value to the buyer and therefore generally reduced to zero.
- Liabilities. Book values for liabilities are typically not altered. However, in situations where the company is in great distress or is filing bankruptcy, the value of its liabilities may be discounted. In addition, for public market debt, the book value may differ from the market value.

Consider each of the assets and liabilities of Paint-Pen. Cash should not be adjusted. In the absence of additional information about the note receivable, one can assume that it will be paid in full since Paint-Pen will continue operating and collecting debts. For purposes of this illustration, accounts receivables and inventories were conjectured to be overstated by 10% and 25%, respectively. Because operations will continue, Paint-Pen is projected to use its prepaid expenses and their value is presumed to be unchanged from the time of the initial payment. Furniture and other equipment, as well as the non-compete covenants, foreign license agreements, and patents are estimated to have values 50% below the corresponding book values. Goodwill is valued at zero. Subtracting the unadjusted total liabilities results in an adjusted book value of \$1,859,151 (see Exhibit 2).

LIQUIDATION VALUE

Liquidation value is an estimate of the worth of the assets and the liabilities assuming that the company will cease its operations and the assets will be auctioned immediately. These assumptions may affect the worth of each type asset and liability.

- Cash and cash equivalents. These assets can quickly and easily be converted to market values, and therefore do not need to be discounted.
- Non-cash tangible assets. The majority of non-cash assets will sell at prices below their replacement values. For example, notes and accounts receivables should be valued below adjusted book value, because it may be difficult to realize these assets if

debtors forestall repaying in the hopes that the business will shut-down without collecting.

- Intangible assets. Patents, licenses, and other similar intangibles would normally be substantially reduced due to the auction conditions. Intangible assets such as goodwill have no value.
- Liabilities. Since debtholders have the highest priority claims to the proceeds, the full book value of the liabilities will have to be paid, if possible.

In the case of Paint-Pen, cash and liabilities are still assumed to be equal to book value. The note receivable and the accounts receivables are discounted 25% and 50%, respectively. Inventories and office equipment are also cut in half. All other assets are projected to have no value. The liquidation value is therefore \$844,080 (see Exhibit 3). These numbers assume an orderly liquidation of the assets. In the case of a forced liquidation, the value for these assets may be considerably lower.

The usefulness of a liquidation valuation is limited since most buyers intend to continue operating a business as an on-going concern. Nevertheless, this valuation provides two insights. First, it sets a floor price below which a seller is unlikely to bid. Second, for a well-managed company that is generally using its assets efficiently, the maximum downside of an opportunity can be calculated by subtracting the liquidation value from the purchase price.

INCOME STATEMENT VALUATIONS

Income statement multiples attempt to value an opportunity by capitalizing its earnings stream. Three sequential steps are necessary to use this valuation technique.

1. Find a company of known enterprise value (equity plus long-term debt after normalizing the current ratio for industry standards) and earnings in the same industry as the target (see Exhibit 4).
2. Calculate the ratio or multiple of value to earnings for the known entity. This ratio captures the market's view of the value of future cash flows from existing businesses and growth opportunities as well as the riskiness of those cash flows.
3. Apply this ratio to the last twelve months earnings (and sometimes the projected next twelve months earnings) results of the target opportunity. This will yield a valuation estimate.

The following income statement items can be substituted for earnings: sales, profit before tax (PBT), profit after tax (PAT), earnings before interest and taxes (EBIT), and earnings before interest, taxes, depreciation and amortization (EBITDA). In most circumstances, it is best to use EBIT or EBITDA since these two measures most closely approximate the true cash generating potential of a business. Sales is rarely used since it is not a measure of profitability. PAT and PBT are affected by the company's financial structure, since they are calculations of earnings after debt interest expenses. Since a buyer may not want to use the same capital structure for her or his acquisition, it is preferable to use income statement items that do not include the interest effects of the existing financial structure.

Profit after tax creates additional complications since tax rates often vary among companies, even within the same industry. EBIT and EBITDA are measures of profitability that are not affected by the existing financial structure or taxes.

EBITDA is a preferable measure to EBIT. Earnings before interest and taxes includes the effects of depreciation and amortization. Accounting practices for depreciation often do not reflect the real utility of the items being depreciated, and therefore distort the cash flow being returned to the equity holders. EBITDA by definition does not account for depreciation and amortization. For some industries, however, accounting principles accurately depict the useful economic life of the depreciable assets. In these industries, the choice between EBIT and EBITDA is arbitrary.

When using income statement multiples, pay close attention for extraordinary expense items which skew historical results and are not expected to recur in the future. For example, previous owners may have paid themselves larger salaries than the new owners anticipate paying themselves. Alternatively, some expenses may be abnormal in relation to industry standards. The statements should be adjusted accordingly to negate these distortions.

In applying this technique, care must be taken on a number of issues. First, a practitioner should look at the multiples of companies in the same industry, of approximately the same size in sales, and with similar performance histories. Second, the ratios need to be applied consistently; never apply EBIT multiples to EBITDA, PBT, or any income statement item other than EBIT. Likewise, always distinguish between trailing and forward looking multiples. Trailing results are best used to determine the value of a company today whereas forward looking results should be used to determine the value of a company in the future. Finally, recognize that many companies are bought and sold outside the industry range.

Comparison companies and their corresponding multiples are often difficult to find, particularly in industries where public companies do not exist. In addition, a practitioner may not have time to find comparables. In these situations, historical multiples can be used (see Exhibit 5 for sample ranges for EBIT and EBITDA).

Turning to Paint-Pen, the term “operating profit” in the income statement is commonly used to indicate either EBIT or EBITDA, depending on the company. In this case, since operating profit is before depreciation, it represents EBITDA of \$1,689,900. Since no comparison information is provided, an appropriate ratio can be found in Exhibit 5. Over the past two years, sales grew 15% and 24%, respectively, but in 1994 revenues increased only by 3%. Furthermore, given the non-glamorous nature of the industry, an average multiple appears appropriate. In this case, an EBITDA multiple of four yields a value of \$6,759,600. To get EBIT, subtract the \$123,010 provision for amortization from EBITDA (“Other income” generally should not be included in EBIT calculations since these earnings do not result from company operations). Using a multiple of five yields a valuation of \$7,834,450 (see Exhibit 6).

DISCOUNTED CASH FLOWS

The theory behind discounted cash flows (DCF) is that the value of a business is equal to the future expected cash flows discounted at a rate that reflects the riskiness of the cash flow stream. Four steps need to be followed to use this valuation technique:

1. projecting the cash flow stream,
2. choosing a discount rate,
3. determining a terminal value for the business at the end of the projections, and
4. applying the discount rate to the projected cash flow stream and the terminal value.

PROJECTING THE CASH FLOW STREAM

“Total free cash flow” is the residual returned to the equity holders after expenditures for maintaining operations (capital expenditures and changes in working capital) and financing the company (interest and taxes). It is calculated as follows:

$$\text{Total Free Cash Flow} = \text{EBITDA} - (\text{Tax Exposure}) - (\text{Changes in Working Capital}) \\ (\text{Required Capital Expenditures})$$

As with income statement multiples, the practitioner should exclude all extraordinary expense items that are not expected to continue into the future from the cash flow calculations. However, any unusual expenses that can be predicted should be included. The number of years that the cash flow stream needs to be projected for depends on the volatility of those cash flows; volatile cash flows should be projected farther than more stable ones in order to allow the practitioner to include as much of the volatility as possible in the model. Five to ten years is a normal range. Realistic assumptions should be made regarding sales growth and margin improvements.

CHOOSING A DISCOUNT RATE

To derive a discount rate, a practitioner may use the weighted average cost of capital (WACC) which accounts for the industry risk, the company-specific risk, and the financial risk (corporate debt) of the target company. The following formula should be used:

$$\text{WACC} = K_E (E/(E + D)) + K_D (D/(E + D)) (1 - t)$$

Where:

- WACC = the weighted average cost of capital
- E = target amount of equity in the company
- D = amount of debt in the company
- K_E = cost of equity
- K_D = cost of debt (interest rates on corporate loans)
- t = corporate tax rate

With the exception of the cost of equity, all these variables can be obtained from publicly and widely available sources. To determine the cost of equity, practitioners may need to use the capital asset pricing model.

$$K_E = R_F + B_L * (R_M - R_F)$$

Where:

K_E = cost of equity

R_F = the risk free rate

B_L = the beta adjusted for the company's projected use of leverage

R_M = the expected market return

Conceptually, the return on equity is a function of the risk free rate of return and a premium (or excess return) for business and financial risk. The interest rate on long-term government treasury bonds can be used for the risk free rate. The market risk premium ($R_M - R_F$) varies over time, but is typically six to eight percentage points.

Beta is a measure of how a company covaries with the market and is approximated using the betas of comparable public companies in the same industry. As best explained in James C. Van Horne's *Financial Management and Policy*, Beta "depicts the sensitivity of the security's excess return to that of the market . . . If it is one, it means that excess return for the stock varies proportionately with excess returns for the market. In other words, the stock has the same unavoidable risk as the market." The larger the beta, the greater the unavoidable risk in relation to the market. Estimates of betas can be obtained from several financial services firms including Merrill Lynch, First Boston, Goldman Sachs, and Value Line Investment Survey.

The beta "is a function of both the business risk and the degree of leverage." If the beta of a comparable company is used to determine the target's beta, and it has a different financial structure (leverage), the use of beta will yield a bias. It is for this reason that the beta should be adjusted to reflect the projected amount of debt the target company will utilize. Thus, the beta for the comparable company should first be un-levered by using the following equation:

$$B_{UL} = B_L / (1 + D/E * (1 - t))$$

Where:

B_{UL} = un-levered beta of the comparable company

B_L = levered beta of the comparable company

E = amount of equity in the comparable company

D = amount of debt in the comparable company

t = corporate tax rate

The un-levered beta can then be re-levered using the parameters of the target company.

$$B_L = B_{UL} (1 + D/E * (1 - t))$$

Where:

B_L = levered beta of the target company

B_{UL} = un-levered beta of the comparable company

E = amount of equity in the target company

D = amount of debt in the target company
 t = corporate tax rate

Due to the difficulty in obtaining representative parameters, it is not always desirable to derive the discount rate. In addition, this calculation might be impossible to derive for industries without comparable public companies. In these cases, there are informal ways to approximate the discount rate. One method is to determine a “required rate of return” - the rate which covers the associated risks of the venture. For example, leveraged buy-outs typically expect returns of 25% to 35% annually. Turn-around situations require an even higher return because of the increased chance of bankruptcy – 30% to 40% is normal. A range of 25% to 35% is appropriate for small, profitable companies like Paint-Pen.

Another mechanism to estimate the discount rate is to use the opportunity cost of capital (or hurdle rate), the return expected if the capital was employed elsewhere. If an individual’s alternative was to invest the money in the market in a mutual fund, then the opportunity cost of capital is the expected market return. Such a discount rate can be calculated by adding the market risk premium (six to eight percentage points) to the current interest rate on long-term government treasury bonds. However, since the opportunity cost of capital reflects the risks of the alternative use (the market return), it is likely not be high enough to include all of the risks associated with the target venture. It is therefore preferable to use the required rate of return.

DERIVING A TERMINAL VALUE

A terminal value is the worth of the enterprise at the end of the projected cash flow stream. There are three standard ways to calculate terminal value. First, the cash flow in the final year can be treated as a perpetuity and therefore divided by the chosen discount rate. Second, an EBITDA multiple can be applied to the final year’s cash flow as already discussed. Third, an appropriate price-earnings multiple can be applied to projected or historic earnings. There is little difference between these methods. For example, calculating a terminal value using an EBITDA multiple of five is the same as dividing the final year’s cash flow by a discount rate of 20%.

DISCOUNTING THE CASH FLOW STREAM

To obtain the net present value of the opportunity, simply discount the stream of future cash flows back to the time of the decision using the chosen discount rate.¹

Another method for discounting cash flows is the adjusted present value (APV) method. APV initially assumes that the capital structure of a venture will consist entirely of equity.

¹ For a more thorough discussion of discounted cash flows, please refer to Chapter 8, Financial Management and Policy, Van Horne, Dr. James C., A Simon & Schuster Company, 1997.

The cash flow stream is therefore projected ignoring the effects of debt, and then discounted using the cost of equity (K_E) with an un-levered beta.

APV accounts for the leverage of a venture by separately projecting the net effects of the interest on the debt. Since interest payments are a pre-tax expense, they reduce the tax burden of a company. The net additional effect of the debt is:

$$\text{Net Effects of Debt} = (\text{Interest Payments}) * (1 - (\text{Tax Rate}))$$

These cash flows from the use of debt are then discounted using the corporate cost of debt (the interest rate on the company's bonds).

The cash flows discounted using the cost of equity plus the discounted net effects of debt equal the total value of the opportunity. In scenarios where the corporate debt to equity ratio is expected to remain constant, WACC, which implicitly accounts for the effects of debt, is preferred. However, in situations where the debt to equity ratio will change over time, APV is more appropriate.²

In the case of Paint-Pen, conservative five year projections are made by estimating that sales will grow 7.5% annually (although they grew substantially in 1996 and 1997, they increased only slightly in 1994). EBITDA and PBT margins are calculated by using historical four-year averages. A 40% tax rate on profit before tax is used. In this case, working capital needs are projected by assuming that the ratio of total current assets to sales will remain the same as in 1997. Historical annual capital expenditure requirements are equal to the differences in furniture, fixtures, machinery, equipment and motor vehicles before depreciation from one year to the next; since 1993, the increase has been no more than 0.3% of sales annually. The results of these calculations account for the difference between EBITDA and total free cash flow. The perpetuity method was used to calculate the terminal value.

The case provides no data about comparable betas. A discount rate of 30% is therefore used. The future stream of free cash flows and the terminal value are then converted to present value terms. The result is a \$3,566,419 valuation for the company at the time of the purchase (see Exhibit 7).

WHICH METHOD TO USE?

Balance sheet techniques provide a quick starting point for the valuation process. These techniques, however, provide the least relevant valuation estimates. Balance sheet valuations are not forward looking and do not appropriately account for risk since they are not based on the expected future performance of the company.

² For a more thorough discussion of the adjusted present value method and its relation to the weighted average cost of capital method, please refer to Principles of Corporate Finance, Brealey & Myers, McGraw Hill, 1997.

Income statement multiples is the valuation technique most frequently used since it is the easiest to use and to communicate. Multiples are simple approximations of discounted cash flows. They are forward-looking in stable environments where future cash flows are expected to be similar to current ones. In addition, since they can be easily communicated and specified ahead of time, they are better for stock transfers in buy/sell agreements.

Income statement multiples, however, have several disadvantages. In a future where the economic and industry conditions are not expected to mirror their historical counterparts, the use of income statement techniques is questionable. In addition, this technique only accounts for the risk of an opportunity to the extent that the comparable company has similar characteristics as the target. Finally, this approach is sometimes viewed as too simple and unsophisticated.

Discounted cash flows are also frequently used. This method accounts for the riskiness of cash flows and the time value of money more accurately than balance sheet or income statement techniques. It is also the most appropriate approach to account for expected changes in the environment. In addition, it best allows the practitioner to conduct sensitivity analysis, therefore providing valuable information about the levers of performance and an understanding about the potential outcomes. Finally, it requires practitioners to explicitly make assumptions that the income statement multiples technique includes implicitly, therefore yielding a more accurate estimate.

The discounted cash flow method, however, is far from perfect. First, future earnings may be difficult to estimate. Second, betas can vary between companies in the same industry and are derived from historical rather than forward-looking data. Finally, in cases where the terminal value is a substantial portion of the total valuation and an EBIT multiple is used to compute the terminal value, the theoretical difference between discounted cash flows and income statement multiples is almost irrelevant.

The three valuation techniques are likely to produce different values for the same opportunity, and as in *Paint-Pen*, can result in a wide range of estimates. Each technique has its strengths and weaknesses, and, when used in conjunction with the others, each adds a unique insight.

Depending on the circumstances of the target opportunity, one approach is likely to be superior to the other two. Balance sheet calculations are most appropriate for industries characterized by high asset intensity. Towing businesses, for example, are valued based on the worth of the company's tow trucks and are commonly sold at prices higher than normal multiples would justify. Income statement techniques are most suitable for moderate growth, service industries with little or no capital equipment. Finally, situations involving very high growth rates or where future operating results are expected to vary considerably from historical results are best valued using discounted cash flows. After gaining familiarity with each technique, a user will develop his or her own judgments as to which methods to use in which situations and the weight to place on each estimate.

OTHER CONSIDERATIONS

Once the basics are grasped, practitioners can begin to fine-tune their valuations. Discounting valuations is common when dealing with issues of illiquidity or control. Non-public companies where investor liquidity is not guaranteed generally trade at a 20% to 35% discount from their public counterparts. Minority shareholders in a company that is majority-owned by another party often get a 10% to 20% discount; conversely, investors sometimes pay a premium for control.

There is a difference between financial and strategic buyers. Financial buyers evaluate an opportunity on a stand-alone basis. A strategic buyer (for example, a large corporation in the same industry as the target opportunity) calculates a valuation based on combining the operations. The strategic buyer generally places a higher value on an opportunity than a financial buyer; the strategic value equals the value to the financial buyer plus the value of the expected synergies from the merger.

Operating metrics, as opposed to historical or projected financial metrics, are also commonly used to value the opportunity. This method applies valuation techniques to an individual customer or a unit of production rather than to the business as a whole. For example, until recently cable companies were bought and sold at a value of \$2,000 per subscriber. Similarly, cement plants often trade based on a price per metric ton of capacity. Such metrics should be seen as complementary to the aggregate financial approaches.

If the acquirer anticipates significant improvement in the operations of a company after the purchase due to a change in management, he or she should not necessarily pay the seller for the additional value that will be created. Thus, valuations are as much based on who is managing the business as in what is being valued.

START-UP VALUATIONS

While the techniques discussed above are appropriate for start-up companies with a limited operating history, only discounted cash flows can be used in a situation when the company does not have an operating history. During the first round of financing when a venture is little more than an idea, venture capitalists will rarely use any of the three valuation methods since there are no historical income statements or balance sheets and projecting cash flows is extremely difficult. Professional investors, such as venture capitalists, have developed two methods to value these opportunities.

- **Comparables.** If an internet company, for example, is seeking financing, capital suppliers will compare it to other internet businesses with a similar business focus that are in a similar stage in the life cycle. Adjustments may be made for the specifics of a company (quality of management team, proprietary technology, etc.). Frequently used sources for comparables are VentureOne (see Exhibit 8) and the San Jose Mercury News.

- Required rates of return. The risk of failure for a start-up is high due to the large number of pitfalls that must be avoided and hurdles that need to be surpassed. The required rate of return is therefore correspondingly high – 50% to 70% is standard. In an analysis of venture capital financings, William E. Wetzel, an authority on entrepreneurship, proposed the following rates of returns for start-ups.

Risk-adjusted cost of venture capital.

Stage (risk)	Description of Stage	Expected Returns
Seed	Capital to prove a concept.	80%
Startup	Capital to complete product development and initial marketing.	60%
First-stage	Capital to initiate full-scale manufacturing and sales	50%
Second-stage	Working capital for the initial expansion of a company that is producing and shipping and has accounts receivables.	40%
Third-stage	Capital for a firm with increasing sales that is at least breaking even.	30%
Bridge	Capital used by a company that will go public in six months to a year.	25%

Instead of using these rates to discount future cash flow streams, risk investors use these rates to project their required returns at some point in the future. In addition, instead of explicitly using a required rate of return, venture capitalists frequently use a different but related metric – the multiple of the initial investment that must be recovered within specified time period. For example, five times the initial investment in four years, which is the same as requiring a 50% rate of return (see Exhibit 9). If an internet company was asking for \$1 million in exchange for 40% of the company (making the total post investment value \$2.5 million), the investor would have to see the potential for the initial investment (\$1 million) to grow to \$5 million in four years (after accounting for equity dilution if more capital was needed in the future). To validate the hurdle rates (or required rates of return) venture capitalists may project the market size and the company's corresponding market share as well as use comparables in the same industry. Alternatively, they may value the company by using the entrepreneur's financial projections (generally after first reducing them or applying a "hair-cut") and using a multiple of EBITDA at some point in the future when the company was profitable and was in a "stable and predictable" state.

Valuing start-up opportunities is far from a perfect science. Furthermore, in addition to providing money, professional risk capital investors provide assistance with strategy, corporate partnerships, management recruiting, and other key needs of a new venture. Their capital may be seen as more valuable to the entrepreneur than capital from sources that can not provide similar help. An entrepreneur therefore may give more equity for the same amount of money to a professional investor than to a less value-added investor.

CONCLUSION

Valuation is a critical skill of any successful investor or entrepreneur. Like many things, learning this art is better done by “trail under fire”; practicing in a sterile setting is rarely equivalent to practicing in a situation where the sale or acquisition of a company is a real possibility.

The price a buyer pays for an opportunity and the value of it should not be confused. Price is as much dependent on the negotiating skills of the parties involved, the dynamics of the sale, and the motivations of the current owner as it is on the valuation of the company. In addition, the structure of the agreement and the timing of the payments are often as important as the price. The abilities to move from a calculated valuation to an agreed price, to negotiate the best possible deal, and to arrange the financial structure are also extremely important to entrepreneurs and investors. These skills are best honed through practice and experience.

Exhibit 1

Paint-Pen, Inc.
Consolidated Balance Sheets

	1996	1995	1994	1993
Assets				
Current assets				
Cash	498,058	878,900	800,440	688,390
Note receivable	623,902	-	-	-
Accounts receivable (less allowance for doubtful collections)	631,070	455,050	357,460	280,910
Inventories	910,790	757,040	590,500	592,610
Prepaid expenses	90,900	61,300	47,000	22,390
Total current assets	2,754,720	2,152,290	1,795,400	1,584,300
Furniture, fixtures, machinery and equipment, motor vehicles	226,690	176,040	153,280	151,480
Less accumulated depreciation	139,440	103,230	96,450	64,370
	87,250	72,810	56,830	87,110
Covenant not to compete, foreign license agreement, patents, etc.	821,800	1,221,800	1,921,800	1,931,800
Less accumulated amortization	245,660	522,640	519,150	359,630
	576,140	699,160	1,402,650	1,572,170
Goodwill	700,000	700,000	-	-
Total assets	4,118,110	3,624,260	3,254,880	3,243,580
<u>Liabilities</u>				
Current liabilities				
Note payable	-	275,000	1,000,000	1,750,000
Accounts payable and accrued liabilities	223,970	195,830	117,810	99,930
Federal income tax payable	712,490	609,660	468,890	325,890
Total current liabilities	936,460	1,080,490	1,586,700	2,175,820
Capital stock and surplus:				
Authorized 1,000 shares of common, no par value - issued and outstanding	2,000	2,000	2,000	2,000
Earned surplus	3,179,650	2,541,770	1,666,180	1,065,760
Total capital stock and surplus	3,181,650	2,543,770	1,668,180	1,067,760
Total Liabilities and net worth	4,118,110	3,624,260	3,254,880	3,243,580

Exhibit 1 (continued)

Paint-Pen, Inc.
Consolidated Statement of Income and Surplus

	1997	1995	1994	1993
Net sales	\$6,883,270	\$5,976,030	\$4,813,500	\$4,665,800
Cost of goods sold	4,319,750	3,599,560	2,881,880	2,895,190
Gross profit	2,563,520	2,376,470	1,931,620	1,770,610
Selling, shipping, general, and administrative expenses	873,620	788,930	676,920	647,880
Operating profit	1,689,900	1,587,540	1,254,700	1,122,730
Other income	19,720	16,140	47,060	64,530
Net income	\$1,709,620	\$1,603,680	\$1,301,760	\$1,187,260
Other charges:				
Provision for amortization	123,010	128,010	169,520	174,520
Interest	6,240	41,210	62,930	91,850
	129,250	169,220	232,450	266,370
Profit before tax	1,580,370	1,434,460	1,069,310	920,890
Provision for federal income tax	712,490	609,660	468,890	325,890
Net profit for the year	\$867,880	\$824,800	\$600,420	\$595,000
Surplus January 1	2,541,770	1,666,180	1,065,760	470,760
Add: Partial disallowance by Treasury Department of amortization of foreign license agreement, patents, etc.		124,520		
		1,790,700		
Deduct: Additional federal income taxes		73,750		
		1,716,950		
Dividends \$230.00 per share	230,000			
Earned Surplus December 31	\$3,179,650	\$2,541,770	\$1,666,180	\$1,065,760

Exhibit 2

Paint-Pen, Inc.
Adjusted Book Value

	1996 Book Value	Adjustment Factor	1996 Adjusted Book Value
Assets			
Current assets			
Cash	\$498,058	100%	\$498,058
Note receivable	623,902	100%	623,902
Accounts receivable (less allowance for doubtful collections)	631,070	90%	567,963
Inventories	910,790	75%	683,093
Prepaid expenses	90,900	100%	90,900
Total current assets	<u>\$2,754,720</u>		<u>\$2,463,916</u>
Furniture, fixtures, machinery and equipment, motor vehicles, net	87,250	50%	43,625
Covenant not to compete, foreign license agreement, patents, etc., net	576,140	50%	288,070
Goodwill	700,000	0%	-
Total assets	<u>\$4,118,100</u>		<u>\$2,795,611</u>
Liabilities			
Current liabilities			
Note payable	-	100%	-
Accounts payable and accrued liabilities	223,970	100%	223,970
Federal income tax payable	712,490	100%	712,490
Total current liabilities	<u>\$936,460</u>		<u>\$936,460</u>
Net worth	\$3,181,650		\$1,859,151

Exhibit 3

Paint-Pen, Inc.
Liquidation Value

	1996 Book Value	Adjustment Factor	1996 Liquidation Value
Assets			
Current assets			
Cash	\$498,058	100%	\$498,058
Note receivable	623,902	75%	467,927
Accounts receivable (less allowance for doubtful collections)	631,070	50%	315,535
Inventories	910,790	50%	455,395
Prepaid expenses	90,900	0%	-
Total current assets	2,754,720		1,736,915
Furniture, fixtures, machinery and equipment, motor vehicles, net	87,250	50%	43,625
Covenant not to compete, foreign license agreement, patents, etc., net	576,140	0%	-
Goodwill	700,000	0%	-
Total assets	4,118,100		1,780,540
Liabilities			
Current liabilities			
Note payable	-	100%	-
Accounts payable and accrued liabilities	223,970	100%	223,970
Federal income tax payable	712,490	100%	712,490
Total current liabilities	936,460		936,460
Net worth	\$3,181,650		\$844,080

Exhibit 4

Multiples for Corporate Acquisitions
Month of July, 1997

Company	SIC Code	Business Description	Enterprise Value* (\$MM)	Multiples				Sales
				EBIT	EBITDA	PAT	PBT	
Norwest Airlines	4512	Air transport	6,649.9	6.3	4.6	12.2	7.5	0.7
Arden Group	5411	Grocery stores	51.5	6.4	NA	9.2	5.5	0.2
Champion Enterprise	1521	Resdnt'l housing	741.9	6.4	5.6	13.8	7.9	0.4
Westerbeke	3519	Engine mfg	9.5	6.6	5.1	11.0	6.5	0.4
Gibson Greetings	2771	Greeting cards	259.0	6.7	2.9	11.5	7.9	0.7
American Filtrona	3569	Ind. machines	137.2	7.0	5.5	10.2	6.5	0.7
Great Eastern Energy	1311	Petro & nat. gas	2.4	7.2	3.0	7.6	7.6	1.1
Insilco	3714	Car parts	419.1	7.3	5.6	4.4	2.9	0.7
Calnetics	3089	Plastic products	26.4	7.3	5.9	14.1	8.0	0.7
Freeport-McMoRan	2874	Fertilizers	1,233.5	7.6	5.4	11.0	9.8	1.4
NY State Gas & Elec	4911	Electric services	3,403.9	7.9	5.5	21.1	13.0	1.7
Firecom	3669	Communications	91.7	8.0	7.6	13.4	11.6	1.2
Stanhope	5199	Nondurables	660.7	8.0	7.2	17.0	9.4	0.8
IBP, Inc.	2011	Meat packing	2,332.4	8.1	6.3	13.1	8.2	0.2
Silicon Storage	3674	Semiconductors	65.2	8.2	5.5	11.6	6.8	0.7
Figgie Int'l	3569	Ind. machines	346.3	8.3	7.0	6.8	13.3	0.9
Great Lakes Chem.	2819	Ind. gases	3,602.6	8.6	7.0	15.2	10.0	1.7
Castle Energy	1311	Petro & nat. gas	100.5	8.6	**	5.1	9.0	1.4
PH Glatfelter	2611	Pulp mills	921.9	8.8	6.6	15.6	9.6	1.6
Centex Const. Prod.	1541	Ind. bldgs	557.0	8.8	7.3	13.3	8.6	2.3
Scope Industries	4953	Refuse systems	58.6	8.9	6.7	3.1	2.3	1.8
Plasti-Line	3993	Signs	57.7	9.1	6.7	17.2	10.9	0.4
Franklin Electric Co.	3621	Electric motors	308.4	9.2	7.4	14.2	9.2	1.0
Schottenstein Homes	1531	Operative builder	245.4	9.7	9.2	15.5	9.5	0.4
SL Industries	3612	Power distribut'n	66.6	9.8	7.1	17.7	11.0	0.6
Orange & Rockland	4911	Electric services	947.7	9.8	7.3	24.5	15.6	1.1
Upper Pen. Energy	4911	Electric services	120.2	9.9	6.8	22.3	14.6	2.0
Amer. Clss. Voyages	4489	Water trans.	707.3	10.4	3.9	71.9	20.2	0.4
Span-Amer. Medical	3842	Orthopedic	15.1	10.9	6.8	14.3	8.9	0.5
Wyle Electronics	5065	Electronic parts	751.3	11.0	9.6	20.3	12.4	0.6
Equitable of Iowa	6311	Life insurance	2,586.8	11.1	10.6	20.6	13.8	2.8
Burlington Northern	4011	Railroads	19,649.6	11.6	8.0	23.1	14.3	2.4
Katz Media Group	7319	Advertising	361.1	12.1	8.5	116.7	41.9	2.0
LADD Furniture	2511	Wood furniture	243.9	12.2	7.1	48.3	30.4	0.5
Mosaix Inc.	3661	Telephone equip.	142.3	12.2	8.3	15.9	10.7	1.2
Int'l Imaging Mat.	3955	Carbon paper	227.2	12.4	8.7	19.4	12.7	2.2
PLM Int'l Inc.	4741	RR car rental	152.4	12.4	6.6	33.5	33.9	3.0
Chips & Technology	3674	Semiconductors	394.4	12.9	11.8	11.8	10.6	2.3
Sysco Corp	5149	Groceries	7,168.3	13.5	10.4	24.3	14.8	0.5

Source: Securities Data Corporation

*Enterprise value equals equity plus long-term debt after normalizing the current ratio for industry standards.

**Not public.

Exhibit 5**Common EBITDA & EBIT Multiples**

EBITDA Multiple	EBIT Multiple	Company Characteristics
3	4	Companies with an unglamorous product line, slow or no growth, and/or low market share
4-6	5-7	A typical range within which a majority of private, non-technology, moderate growth companies sell
6+	7+	Appropriate only for opportunities the buyer considers exceptional or for businesses growing faster than 10% annually.

Note: EBITDA multiples are lower than EBIT multiples because depreciation is not reflected in the former. Typical ranges for other income statement multiples are one to two times sales, six to eight times profit before tax, and eight to ten times profit after tax.

Exhibit 6

Paint-Pen, Inc.
Income Statement Multiples Valuations

	1996 Income Statement	Multiple	Valuation
EBITDA*	\$1,689,900	4X	\$6,759,600
Subtract: provision for amortization	<u>\$123,010</u>		
EBIT	\$1,566,890	5X	\$7,834,450

*Reported as "Operating profit" in Consolidated Statement of Income and Surplus.

Exhibit 7

Paint-Pen, Inc.
Discounted Cash Flow Valuation

	Actual	Projections				
	1997	1998	1999	2000	2001	2002
Income statement						
Net sales	6,883,270	7,399,515	7,954,479	8,551,065	9,192,395	9,881,824
EBITDA*	1,689,900	1,816,643	1,952,891	2,099,357	2,256,809	2,426,070
Profit before tax	1,580,370	1,698,898	1,826,315	1,963,289	2,110,535	2,268,826
Provisions for tax	712,490	679,559	730,526	785,315	844,214	907,530
Net profit for the year	867,880	1,019,339	1,095,789	1,177,973	1,266,321	1,361,295
Balance sheet						
Total current assets**	2,754,720	2,959,806	3,181,792	3,420,426	3,676,958	3,952,730
Furniture, fixtures, etc.***	226,690	248,889	272,752	298,405	325,982	355,628
Change in:						
Total current assets		205,086	221,985	238,634	256,532	275,772
Furniture, fixtures, etc.		22,199	23,863	25,653	27,577	29,645
Present value analysis						
Total free cash flow****		909,799	976,516	1,049,754	1,128,486	1,213,122
Terminal value						4,043,742
Total future cash flows		909,799	976,516	1,049,754	1,128,486	5,256,864
Present value of cash flows		699,845	577,820	477,813	395,114	1,415,826
Net present value	3,566,419					

*Reported as "Operating profit" in Consolidated Statement of Income and Surplus.

**Assumes ratio of total current assets to sales remains constant at 0.40.

***Assumes annual increase is 0.3% of sales.

****Total free cash flow equals EBITDA less provisions for tax, change in total current assets, and change in furniture, fixtures, etc.

Exhibit 8

Start-up Valuation Comparables

Company	Round of Financing	Date	Amount Raised (\$MM)	Post-\$ Valuation (\$MM)	Company Stage
Agile Software	Seed	6/1/95	1.0	2.1	Start-up
Business Matters	2nd	6/30/95	2.5	7.8	Product Dev't
Coda Music Technologies	IPO	6/29/95	6.0	24.7	Shipping Product
Continuous Software	3rd	5/25/95	1.8	26.0	Shipping Product
Datalogix International	IPO	6/15/95	56.1	174.8	Profitable
Dendrite International	IPO	6/29/95	37.7	153.0	Profitable
Dynasty Technologies	2nd	2/17/95	5.0	12.0	Shipping Product
Expert Software	IPO	4/11/95	32.4	83.1	Profitable
Flexi International	2nd	1/19/95	2.0	18.1	Shipping Product
Framework Technologies	1st	5/09/95	3.3	8.1	Product Dev't
Hands On Technology	3rd	6/20/95	1.8	6.2	Product Dev't
Illustra	3rd	5/12/95	7.0	43.0	Shipping Product
Ipsys Software	2nd	2/17/95	1.0	11.0	Profitable
Kapre Software	4th	2/24/95	3.1	14.0	Product Dev't
Light Source	3rd	1/09/95	2.5	20.0	Shipping Product
LV Software	1st	1/15/95	1.0	6.7	Product Dev't
LV Software	2nd	7/2/95	1.6	8.0	Product Dev't
Marketplace Info System	2nd	5/1/95	1.0	14.5	Shipping Product
Netsys Technologies	1st	2/1/95	3.2	13.2	Product Dev't
Open Environment	IPO	4/13/95	41.2	103.3	Profitable
Open Solutions	3rd	3/1/95	0.5	10.5	Product Dev't
Penergy	2nd	4/18/95	0.7	9.9	Product Dev't
Portable Software	2nd	7/14/95	3.0	10.2	Shipping Product
Regenisys Software	2nd	3/3/95	3.0	4.3	Product in Beta
Spyglass	IPO	7/1/95	34.0	85.1	Shipping Product
Starware	1st	2/28/95	2.0	10.0	Shipping Product
Telops Management	Seed	4/15/95	1.5	3.5	Start-up
Travelnet	3rd	6/28/95	6.0	22.8	Product in Beta
Unison Software	IPO	7/20/95	20.8	66.7	Profitable
Verisign	1st	4/20/95	5.0	10.0	Shipping Product
Vividus	2nd	1/26/96	0.7	4.0	Shipping Product
WorldTalk	3rd	3/3/95	3.7	15.0	Shipping Product
Xaos Tools	2nd	2/1/95	1.4	24.8	Shipping Product

Source: VentureOne

Exhibit 9**Rate of Return Calculations**

Years	Multiple of Original Investment									
	1X	2X	3X	4X	5X	6X	7X	8X	9X	10X
1	0%	100%	200%	300%	400%	500%	600%	700%	800%	900%
2	0%	41%	73%	100%	124%	145%	165%	183%	200%	216%
3	0%	26%	44%	59%	71%	82%	91%	100%	108%	115%
4	0%	19%	32%	41%	50%	57%	63%	68%	73%	78%
5	0%	15%	25%	32%	38%	43%	48%	52%	55%	58%
6	0%	12%	20%	26%	31%	35%	38%	41%	44%	47%
7	0%	10%	17%	22%	26%	29%	32%	35%	37%	39%
8	0%	9%	15%	19%	22%	25%	28%	30%	32%	33%
9	0%	8%	13%	17%	20%	22%	24%	26%	28%	29%
10	0%	7%	12%	15%	17%	20%	21%	23%	25%	26%